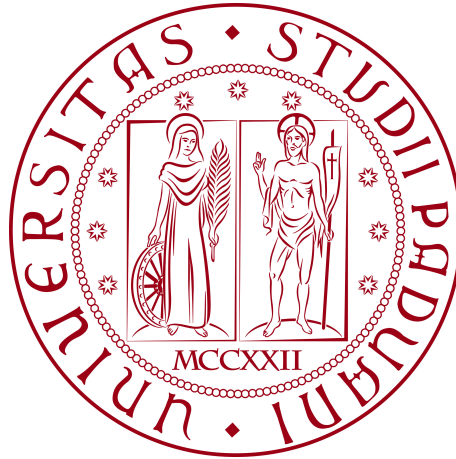


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“We can only see a short distance ahead, but we can see plenty there that needs to be done.” — Alan Turing

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This thesis marks the conclusion of an important chapter of my academic journey and, at the same time, the culmination of a path that has allowed me to explore Artificial Intelligence in greater depth. During these years, I have had the opportunity to learn not only about algorithms, models, and technologies, but also about the broader questions surrounding the development and use of intelligent systems. The rapid evolution of Artificial Intelligence

makes it difficult to predict where this field will lead in the future, but it also makes clear that there is still much to discover, understand, and build.

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Abstract

This thesis investigates the use of Structured Logic Tensor Networks (LTNs) for neuro-symbolic learning. The goal is to combine neural learning with explicit background logical knowledge. The proposed approach addresses an AI classification task with limited and heterogeneous training data. Entities and concepts are represented through learnable continuous embeddings. Logical relations and domain constraints are expressed using first-order logic. These logical formulas are converted into differentiable constraints using fuzzy logic semantics. The model jointly learns from labeled examples and the available background knowledge. A structured representation is used to organize concepts, relations, and logical dependencies. This allows domain knowledge to guide the learning process rather than relying only on observed data. The approach is evaluated against conventional neural network baselines. Experiments investigate both predictive accuracy and logical consistency of the learned model. Additional experiments analyze the effect of increasing or reducing the amount of background knowledge. The results examine whether logical constraints improve generalization, particularly with limited training data. The thesis also discusses computational and modeling challenges associated with differentiable logical reasoning. Overall, the study demonstrates the potential of Structured LTNs for integrating learning and reasoning in AI systems.

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Chapter 1

Introduction

Artificial intelligence systems are increasingly required not only to learn from data, but also to reason about the knowledge acquired from their environment. Modern machine learning methods have demonstrated remarkable capabilities in processing complex and high-dimensional data, including images, natural language, and other forms of unstructured information. However, learning directly from data does not by itself provide an explicit representation of knowledge or a principled mechanism for expressing logical constraints. Conversely, symbolic artificial intelligence provides powerful formalisms for representing and manipulating structured knowledge, but traditionally relies on explicitly defined symbols and rules and therefore faces difficulties when dealing with noisy, incomplete, or high-dimensional data. The integration of learning and reasoning has consequently become a central topic in neurosymbolic artificial intelligence [1]. Within this context, Logic Tensor Networks (LTNs) provide a framework for integrating first-order logic with differentiable neural computation. LTN introduces a many-valued, end-to-end differentiable logical language, called Real Logic, in which the elements of a first-order logic signature are grounded onto data through neural computational graphs. Logical formulas are assigned truth values in the interval $[0, 1]$, allowing logical constraints to participate directly in gradient-based learning [1]. This provides a common framework in which learning from data and reasoning with background knowledge can be expressed within the same formalism. More recently, the Structural Logic Tensor Networks (sLTN) framework extends the original LTN formulation to explicitly represent structural organization in data. While conventional LTN is primarily formulated for domains consisting of collections of individuals, many real-world datasets possess additional dimensions such as temporal order, sequence position, or graph structure. sLTN introduces

structural dimensions, structural variables, and structural relations as explicit components of the logical language, allowing such organization to be represented and reasoned about at the logical level [2]. This thesis is situated within this transition from general differentiable logical reasoning over collections of individuals towards logical reasoning over structured data.

1.1 Logic Tensor Networks

Logic Tensor Networks were introduced as a neurosymbolic framework designed to combine the representational capabilities of first-order logic with the learning capabilities of neural networks. The central idea is to interpret the symbols of a logical language using differentiable tensorial representations, so that logical knowledge can be incorporated directly into a machine learning system [1]. Traditional machine learning approaches generally represent knowledge implicitly through numerical parameters learned from data. Such approaches are particularly effective when large amounts of labelled or unlabelled data are available, but the resulting representations do not necessarily expose the logical relationships that hold between concepts. Symbolic approaches, on the other hand, represent knowledge explicitly using concepts, relations, and logical rules. This makes symbolic representations suitable for reasoning and knowledge manipulation, but classical symbolic systems are generally less suited to directly processing raw, noisy, or high-dimensional observations [1]. Neurosymbolic approaches attempt to bridge this distinction by combining symbolic representations with sub-symbolic learning. In this setting, symbolic knowledge can provide constraints and relationships that guide learning, while neural models provide mechanisms for learning representations from data. A fundamental issue in such systems is therefore the grounding of symbolic entities into concrete data representations. LTN addresses this problem by assigning differentiable tensor representations and functions to the elements of a first-order logical language [1]. The LTN framework is based on Real Logic, a many-valued and differentiable extension of first-order logic. In this setting, logical formulas are not restricted to the classical truth values of true and false. Instead, their truth values are represented numerically in the interval $[0, 1]$. Logical connectives and quantifiers are implemented using differen-

differentiable fuzzy-logic operations, allowing the satisfaction of logical formulas to be optimized using gradient-based methods [1]. This formulation leads to an important reinterpretation of learning. Rather than defining learning solely as the minimization of a task-specific numerical loss, LTN allows the learning problem to be expressed through a collection of logical formulas whose degree of satisfaction is optimized. The parameters of the underlying neural computational graphs can consequently be learned so as to maximize the overall truth value of the specified knowledge [1]. The resulting framework can express a variety of learning and reasoning tasks. The original LTN work demonstrates applications including data clustering, multi-label classification, relational learning, query answering, semi-supervised learning, regression, and embedding learning [1]. The common feature of these tasks is that data-driven learning and logical constraints can be expressed within a single differentiable framework. An LTN model can therefore be viewed as consisting of two interacting components. On the one hand, neural computation provides continuous representations and differentiable mappings from data to these representations. On the other hand, the logical layer specifies relationships and constraints between the represented entities. The grounding connects these two levels by mapping logical terms and predicates to differentiable tensor operations [1]. This combination is particularly relevant when background knowledge is available. Instead of treating the available knowledge as an external post-processing mechanism, LTN incorporates it directly into the learning objective. The logical structure can consequently influence the learned representations while the neural components provide the flexibility required to learn from numerical data [1].

1.2 From Flat Domains to Structured Data

Despite the generality of the LTN framework, its original formulation is primarily oriented towards domains that can be described as collections of individuals and relations between them. This setting is appropriate for many relational learning problems, where entities can be represented independently and logical predicates describe their properties or relationships. However, a large class of modern datasets contains an additional structural organization that cannot be naturally described only through relations between independent individuals [2]. Examples include temporal sequences, where observations are associated with ordered time steps; sequential data, where the position of an element within a sequence is significant; and graphs, where the connectivity between nodes provides essential information about the domain. In these settings, the position or structural context of an observation is itself meaningful. Treating such information merely as an additional property of an individual does not necessarily provide a natural representation of the underlying structure [2]. Consider a temporal dataset represented as a tensor. One axis may correspond to different individuals or samples, while another axis represents time. The temporal axis has a meaning that is different from the identity of the individuals contained in the tensor. Similarly, in a sequence, the position of an element is not simply another individual in the domain: it represents its location within an ordered structure. In a graph, the organization of nodes and their connectivity introduces another form of structure that is not captured solely by treating nodes as an unordered collection [2]. This distinction motivates the introduction of explicit structural information into the logical language. The sLTN framework addresses this problem by extending LTN with structural dimensions that represent named dimensions of structured objects. Examples of structural dimensions include time steps, sequence positions, and graph nodes [2]. Structural dimensions are therefore conceptually different from ordinary logical sorts and individuals. An ordinary first-order variable ranges over individuals in a domain, whereas a structural variable ranges over positions associated with a structural dimension. This distinction allows logical formulas to refer explicitly to the organization of data rather than encoding structural information indirectly [2]. The introduction of structural dimensions also makes it possible to define structural relations. A structural

relation can describe relationships between positions within a structure. For example, a relation such as

$$\text{next}(t, t')$$

can express that t' is the position immediately following t in a temporal or sequential dimension. Such relations can then be used inside logical formulas to express constraints involving neighbouring or otherwise structurally related positions [2]. This extension is important because structural relationships can influence the interpretation of a formula. A constraint may not simply state that a property holds for an individual, but that it holds at a particular structural position, or that properties at two positions are related according to the structure of the underlying data. sLTN consequently provides a way to express temporal, sequential, and relational constraints directly within the logical language [2].

1.3 Structural Logic Tensor Networks

Structural Logic Tensor Networks extend the LTN framework by making structural dimensions first-class elements of the logical language. The framework introduces an explicit distinction between the logical domain of individuals and the structural organization associated with tensors. This allows logical expressions to refer to positions within structured data while retaining the differentiable fuzzy semantics of LTN [2]. The formal language of sLTN includes ordinary first-order elements such as sorts, constants, variables, functions, and predicates, together with structural elements such as structural variables and structural relations. Structural dimensions provide the link between the logical description and the axes of the underlying tensors [2]. A structural dimension can be understood as a named axis associated with a particular type of organization in the data. For example, a dimension called *time* may identify the temporal axis of a tensor, while another dimension may represent positions in a sequence. The structural dimension is not merely metadata: it participates in the interpretation of logical expressions by determining how tensorial data are aligned, selected, masked, or aggregated [2]. Structural variables are used to quantify over the positions of

these dimensions. In contrast to ordinary first-order variables, which range over individuals, structural variables range over indices associated with structural dimensions. This distinction allows formulas to make explicit statements about the location of data within a structured object [2]. Structural relations provide a further level of expressiveness. They allow relationships between structural positions to be represented explicitly. For example, a temporal relation can identify consecutive time steps, while other structural relations can describe more general dependencies between positions. These relations can be Boolean or fuzzy and can therefore be integrated into the differentiable semantics of the framework [2]. An important consequence of this design is that structural information becomes part of the logical theory rather than being handled exclusively by the underlying neural architecture. The logical language can therefore express constraints whose meaning depends directly on the organization of the data. This is particularly relevant for problems in which temporal or sequential dependencies are part of the knowledge that should guide learning [2].

The sLTN framework retains the central principle of LTN: logical expressions are interpreted through differentiable tensor operations. Thus, introducing structural dimensions does not replace the original tensorial semantics but extends it to account for additional axes of organization. The resulting framework can align logical expressions with tensor dimensions and use structural information when performing selection, masking, or aggregation operations [2]. An important theoretical property of sLTN is that the original LTN framework can be recovered as a special case. When no structural dimensions are introduced, the structural extension does not add additional structure to the interpretation, and the framework reduces to the original LTN setting [2]. This establishes a direct relationship between the two formulations rather than defining sLTN as an independent logical framework.

1.4 Syntax, Semantics, and Tensorial Interpretation

A central aspect of sLTN is the explicit separation between the syntactic and semantic levels of the framework. The language first defines the symbols and expressions that are permitted by the theory. These expressions are then parsed and validated before being interpreted over tensorial data [2]. The signature specifies the vocabulary available to a theory. It contains the sorts, dimensions, constants, variables, functions, predicates, and structural relations required to construct formulas. This explicit signature provides a formal description of the entities and relationships that can appear in a logical theory [2]. The syntactic layer is followed by formula parsing and validation. This separation is important because it distinguishes the logical structure of an expression from the numerical representation used to evaluate it. A formula can therefore be treated as a symbolic object before being mapped to tensor operations by the interpretation mechanism [2]. At the semantic level, the symbols of the logical language are assigned tensorial interpretations. Constants and variables are grounded into tensor representations, while predicates and functions are interpreted through differentiable operations. Logical connectives and quantifiers are then evaluated using fuzzy semantics, producing numerical truth values [1, 2]. Structural dimensions introduce an additional component into this interpretation. A structural dimension determines an axis of the tensorial representation, while structural variables identify positions along that axis. Structural relations determine relationships between such positions. The interpretation can consequently use the structural information to align tensors and to perform operations that respect the organization encoded by the data [2]. This tensorial interpretation is essential for maintaining differentiability. Logical reasoning is not implemented as a separate symbolic process disconnected from learning. Instead, formulas are translated into differentiable numerical operations whose outputs can contribute to an optimization objective. As in LTN, the truth value of a formula can therefore influence the learning of the parameters involved in its grounding [1]. The separation between syntax and semantics also provides a modular view of the framework. The logical theory specifies what can be expressed, while the interpretation specifies how those expressions are connected to the underlying data. This distinction becomes particularly useful for structured data because the same logical concepts

can be associated with different tensorial organizations through their structural dimensions [2].

1.5 Motivation and Objectives

The motivation for studying structured extensions of Logic Tensor Networks follows from the increasing importance of structured and multidimensional data in machine learning. Temporal sequences, ordered observations, and graph-structured information contain relationships that are intrinsic to their organization. Representing such information within a differentiable logical framework requires mechanisms capable of referring explicitly to positions and structural relationships [2]. The original LTN framework provides the fundamental combination of first-order logic, fuzzy semantics, tensorial grounding, and differentiable learning [1]. sLTN extends these principles by introducing structural dimensions and relations into the logical language, thereby providing a mechanism for expressing constraints over structured data [2]. From this perspective, the objective is not to replace neural computation with symbolic reasoning. Rather, the objective is to provide a common formal language in which numerical representations and logical constraints can coexist. Neural components remain responsible for learning continuous representations, while logical formulas provide explicit knowledge and structural constraints that can be incorporated into the learning process [1, 2]. The structural extension is particularly relevant when the structure itself carries semantic information. For example, in temporal data, the relationship between two observations may depend on their relative positions in time. In sequential data, the position of an element may determine its relationship with neighbouring elements. In graph data, connectivity determines which entities are directly related. Structural dimensions and relations provide explicit constructs for representing these dependencies [2]. Another important objective is to preserve the differentiable nature of LTN. Introducing structural information should not require abandoning gradient-based optimization. sLTN therefore maintains the tensorial and fuzzy semantics of LTN while extending them to structured dimensions [2]. The resulting perspective can be summarized as the integration of three components: logical knowledge, differentiable representations, and structural organization. Logical knowledge de-

scribes properties and relationships that should hold; differentiable representations connect these concepts to numerical data; and structural dimensions describe how data are organized within tensors. Together, these components provide the conceptual basis for reasoning over structured data using differentiable logical constraints.

Chapter 2

Conclusions and future research

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Articles and papers

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